

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.4: Independence, bases, and rank

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Linear independence

Activity

In the activity we showed that the set

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}$$

is a subspace. In the activity, we showed that the set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a spanning set for W . The set

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$$

also spans W . But why is S ‘better’?

It uses fewer vectors to span the same space.
Which vectors in T are redundant?

$$\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

So T spans W , but it contains repeated information.

A subspace U has many different spanning sets. We use the concept of *linear independence* to determine when sets of spanning vectors are the ‘most efficient’. In data terms, independence asks whether one feature direction is already built from the others.

Definition: Linear Independence A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is said to be

linearly independent

, or simply

independent

if the only solution to the equation $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$ is trivial, i.e., $t_1 = \dots = t_k = 0$.

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is

linearly dependent

if it is *not* linearly independent, i.e.,

if there exists scalars $t_1, \dots, t_k$, not all zero, such that $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.

Activity

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ -2 \end{bmatrix}$$

are linearly independent.

We need to determine the solutions to the equation $t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + t_3\mathbf{v}_3 = \mathbf{0}$. Expanding out this equation gives the system

$$\begin{aligned} 3t_1 + t_2 - t_3 &= 0 \\ 2t_1 + 3t_2 + 4t_3 &= 0 \\ 2t_1 + 0t_2 - 2t_3 &= 0. \end{aligned}$$

We solve this homogeneous system of linear equations using the Gaussian elimination algorithm. First we create the augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 2 & 3 & 4 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right]$$

We reduce the matrix using row operations to

row echelon form:

$$\begin{bmatrix} 3 & 1 & -1 & | & 0 \\ 2 & 3 & 4 & | & 0 \\ 2 & 0 & -2 & | & 0 \end{bmatrix}$$
$$\rightarrow \begin{bmatrix} 3 & 1 & -1 & | & 0 \\ 0 & \frac{7}{3} & \frac{14}{3} & | & 0 \\ 0 & -\frac{2}{3} & -\frac{4}{3} & | & 0 \end{bmatrix}$$
$$\rightarrow \begin{bmatrix} 3 & 1 & -1 & | & 0 \\ 0 & \frac{7}{3} & \frac{14}{3} & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

There is one free variable corresponding to this matrix, which implies that there is one basic solution to the equation, and thus the system has a nontrivial solution. Thus the vectors are *not* linearly independent.

The last example demonstrates an algorithm to verify whether a given set of vectors is linearly independent.

To verify if a set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are linearly independent:

1. Consider the system of linear equations $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.
2. Solve the resulting linear homogeneous system.

3. If there is a nontrivial solution to the equation, then the vectors are linearly dependent. Otherwise, the vectors are linearly independent.

In matrix form: put the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as the columns of a matrix M . Then $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is independent exactly when $M\mathbf{c} = \mathbf{0}$ has only the trivial solution, equivalently when every column of M is a pivot column.

Theorem If $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is a linearly independent set of vectors in $\mathbb{R}^n$, then every vector in their span $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ has a unique representation as a linear combination of the $\mathbf{x}_i$.

Proof Let $\mathbf{x}$ be an element of $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$. If the equation

$$\mathbf{x} = a_1\mathbf{x}_1 + \dots + a_k\mathbf{x}_k$$

had another solution, i.e.,

$$\mathbf{x} = b_1\mathbf{x}_1 + \dots + b_k\mathbf{x}_k,$$

then subtracting one equation from the other gives that

$$(a_1 - b_1)\mathbf{x}_1 + \dots + (a_k - b_k)\mathbf{x}_k = \mathbf{0}.$$

Thus the coefficients $a_1 - b_1, \dots, a_k - b_k$ give a solution to the equation

$$t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}.$$

But the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly independent, and so this equation has only a trivial solution. Thus $a_1 - b_1, \dots, a_k - b_k = 0$, i.e., $a_1 = b_1, \dots, a_k = b_k$, proving that the representation of $\mathbf{x}$ as a linear combination is unique.

Activity

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -3 \\ -1 \\ -1 \end{bmatrix}$$

are linearly independent.

We must check if the equation

$$t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2 + t_3 \mathbf{v}_3 = \mathbf{0}$$

has nontrivial solutions. Expanding out this equation gives the system

$$t_1 + t_2 = 0$$

$$2t_1 - 2t_2 - 3t_3 = 0$$

$$-t_3 = 0$$

$$t_1 - t_3 = 0$$

We could use row-reduction here, though in this case a more ad-hoc approach is more efficient. The third equation gives $t_3 = 0$, and once this is deduced the fourth equation gives $t_1 = 0$. Then

the first equation gives $t_2 = 0$. Thus the only solution to the equation is the trivial solution, and so the vectors are linearly independent.

Activity

Is the set $S = \{0\}$ linearly independent?

Does the equation $t0 = 0$ have a nontrivial solution?

Fact The nonzero rows of a matrix in row echelon form are linearly independent.

Basis and dimension

In geometry, we often describe lines as being ‘1-dimensional’, and planes as being ‘2-dimensional’. We can describe the notion of dimension more precisely using linear algebra.

Theorem: A Fundamental Theorem Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, and if U contains k linearly independent vectors, then $k \leq m$.

Activity

Is it possible to have 4 linearly independent vectors in $\mathbb{R}^3$?

It is not possible. $\mathbb{R}^3$ is spanned by 3 vectors (for example, the standard basis vectors), so by A Fundamental Theorem, any linearly independent subset of $\mathbb{R}^3$ must contain at most 3 vectors.

Definition: Bases of Vector Spaces If U is a subspace of $\mathbb{R}^n$, a set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ in U is called a **basis** of U if it satisfies the following two conditions:

- $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ is *linearly independent*.
- $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$.

Theorem: The Invariance Theorem If $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ are both *bases* for a subspace U of $\mathbb{R}^n$, then $m = k$.

The Invariance Theorem says that, even though a subspace has many different bases, all bases must have the *same number* of vectors.

Definition: Dimensions of Vector Spaces The **dimension** of a non-zero subspace U of $\mathbb{R}^n$ is the *number* of vectors in a *basis* for U . We often write $\dim(U)$ for the dimension of U , and if $\dim(U) = d$, we say U is d -dimensional. By convention, the dimension of the trivial subspace $\{0\}$ is *zero*.

- **Term:** spanning set
Meaning: enough directions to generate the whole subspace
- **Term:** independent set

Meaning: no direction is redundant

- **Term:** basis

Meaning: just enough directions: spanning and independent

- **Term:** dimension

Meaning: number of directions in any basis

Activity

What is the dimension of $\mathbb{R}^3$?

The standard basis for $\mathbb{R}^3$ is a basis, i.e., the set of three vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ defined in Standard basis is linearly independent and spans $\mathbb{R}^3$. Thus $\mathbb{R}^3$ has dimension 3.

Activity

What is the dimension of $\mathbb{R}^4$?

The standard basis $\mathbf{e}_1, \dots, \mathbf{e}_4$ for $\mathbb{R}^4$ is a basis, and is a set of four vectors, so $\dim(\mathbb{R}^4) = 4$.

Activity

What is the dimension of $\mathbb{R}^n$?

The standard basis is a basis for $\mathbb{R}^n$, so $\dim(\mathbb{R}^n) = n$.

Activity

Find the dimension of the space

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}.$$

The set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a basis for W since S is linearly independent (neither vector in S is the scalar multiple of the other). In the solution to the activity, we showed that these vectors span W , so S is

a basis for W . Since S contains two vectors, $\dim(W) = 2$.

Fact Let U be a non-zero subspace of $\mathbb{R}^n$. Then

- U has a basis and $\dim(U) \leq n$.
- Any independent set in U can be enlarged (by adding some subset of vectors in any fixed basis) to obtain a basis of U .
- Any spanning set for U contains a subset of vectors which form a basis for U .

Theorem Let U be a subspace of $\mathbb{R}^n$ where $\dim(U) = m$ and let $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be a set of m vectors in U . Then B is linearly independent if and only if B spans U .

Proof Suppose B is linearly independent and $|B| = m = \dim(U)$. By A Fundamental Theorem, one cannot add any new vectors to the set B and remain linearly independent. But the referenced item says that if B is not already a spanning set for U , one can add new vectors to B from U , and remain linearly independent. Thus B must be a basis.

Conversely, if B spans U , then by the referenced item we can choose a subset of B which is a basis for U . But any proper subset of B contains fewer than m vectors, and thus by The Invariance Theorem, cannot be a basis for U . Thus B is the only subset of itself that can be a basis, and because B must contain a basis, the set B must itself be a basis.

Theorem Let U and W be subspaces of $\mathbb{R}^n$ with $U \subseteq W$. Then:

- $\dim(U) \leq \dim(W)$.
- If $\dim(U) = \dim(W)$, then $U = W$.

Activity

If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$, show that $\{\mathbf{v}, \mathbf{w}\}$ is dependent if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel, i.e., if one vector is a scalar multiple of the other.

If $\mathbf{v}$ and $\mathbf{w}$ are parallel, then one is a scalar multiple of the other, say $\mathbf{v} = a\mathbf{w}$ for some scalar a . Then $1\mathbf{v} - a\mathbf{w} = \mathbf{0}$, which is a nontrivial linear combination of $\mathbf{v}$ and $\mathbf{w}$, so $\{\mathbf{v}, \mathbf{w}\}$ is dependent. Conversely, if $\{\mathbf{v}, \mathbf{w}\}$ is dependent, then $s\mathbf{v} + t\mathbf{w} = \mathbf{0}$ for some scalars s and t , where $s \neq 0$ or $t \neq 0$. If, say, $s \neq 0$, then $\mathbf{v} = -t/s\mathbf{w}$, so $\mathbf{v}$ and $\mathbf{w}$ are parallel. A similar argument works if $t \neq 0$.

Activity

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are non-zero vectors in $\mathbb{R}^n$, where $\{\mathbf{v}, \mathbf{w}\}$ is independent. Show that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent if and only if $\mathbf{u}$ is not in the subspace $M = \text{span}\{\mathbf{v}, \mathbf{w}\}$. See textbook figure fig-5-2-independent and textbook figure fig-5-2-dependent.

If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent, suppose that $\mathbf{u}$ is in $M = \text{span}\{\mathbf{v}, \mathbf{w}\}$, say, that $\mathbf{u} = a\mathbf{v} + b\mathbf{w}$ for some scalars a and b . Then $1\mathbf{u} - a\mathbf{v} - b\mathbf{w} = \mathbf{0}$. Thus a nontrivial linear combination of $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ is equal to zero, contradicting the independence of $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.

On the other hand, suppose $\mathbf{u}$ is not in M . We need to show that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent. If $r\mathbf{u} + s\mathbf{v} + t\mathbf{w} = \mathbf{0}$, where r, s , and t are not all zero, then $r \neq 0$, for otherwise $s\mathbf{v} + t\mathbf{w} = \mathbf{0}$, which contradicts the independence of $\{\mathbf{v}, \mathbf{w}\}$. But then $\mathbf{u} = -s/r\mathbf{v} - t/r\mathbf{w}$, which contradicts that $\mathbf{u}$ is not in M .

Note More generally, if $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are a linearly independent set of vectors in $\mathbb{R}^n$, and $\mathbf{x}_{k+1}$ is another vector in $\mathbb{R}^n$, then $\{\mathbf{x}_1, \dots, \mathbf{x}_{k+1}\}$ is a linearly independent set if and only if $\mathbf{x}_{k+1}$ is not contained in the span of the set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

Activity

Find a basis for $\mathbb{R}^4$ that contains the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix}.$$

Since $\mathbb{R}^4$ is four dimensional, we must find two vectors $\mathbf{w}_1$ and $\mathbf{w}_2$ such that the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}_1, \mathbf{w}_2\}$ is linearly independent. The referenced item states that we can choose these vectors from any basis, so we might as well choose them from the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4\}$.

One strategy is to choose one vector $\mathbf{e}_i$ not contained in the span of the set $\{\mathbf{v}_1, \mathbf{v}_2\}$, and then choose a vector $\mathbf{e}_j$ not contained in the span of the set $\{\mathbf{e}_i, \mathbf{v}_1, \mathbf{v}_2\}$. By the referenced item, the set $\{\mathbf{e}_j, \mathbf{e}_i, \mathbf{v}_1, \mathbf{v}_2\}$ is then linearly independent. The vector $\mathbf{e}_2$ is not contained in the span of the set $\{\mathbf{v}_1, \mathbf{v}_2\}$, since the second entry in any linear combination of the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is equal to

zero, and thus the vector cannot be equal to $\mathbf{e}_2$. We also claim that the vector $\mathbf{e}_1$ is not contained in the span of the set $\{\mathbf{e}_2, \mathbf{v}_1, \mathbf{v}_2\}$. If

$$\mathbf{e}_1 = a\mathbf{e}_2 + b\mathbf{v}_1 + c\mathbf{v}_2,$$

then checking each entry, we get the system of linear equations

$$1 = b$$

$$0 = a$$

$$0 = 2b + 3c$$

$$0 = 3c$$

The first, third, and fourth equations immediately contradict one another, so such a system cannot be solved.

Thus $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{v}_1, \mathbf{v}_2\}$ is a basis for $\mathbb{R}^4$.

Bases and coordinate systems

One way to think of a basis is as providing a coordinate system for a subspace. The uniquely determined coefficients in the linear combination

of a vector in the subspace are the coordinates.

Definition: Ordered Bases of a Subspace Let $S = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of the subspace U . We call the tuple $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ an **ordered basis** of U .

As an example, $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{e}_2, \mathbf{e}_1)$ are two different ordered bases of $\mathbb{R}^2$. The two sets $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$, and so describe the same bases of $\mathbb{R}^2$.

Note Many Linear Algebra textbooks, including *Linear Algebra With Applications*, *Nicholson* write ordered bases using set notation $\{\dots\}$, despite the order in which the basis elements appear being important. We use tuple notation $(\dots)$ to emphasize the importance of the order.

Given an ordered basis B for an m -dimensional subspace U of $\mathbb{R}^n$, and any vector $\mathbf{v} \in U$, we can describe its ‘coordinates’ with respect to the basis B , identifying $\mathbf{v}$ with a vector in $\mathbb{R}^m$.

Definition: Coordinate Vectors With Respect to a Basis Let U be an m -dimensional subspace of $\mathbb{R}^n$, and let $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ be an ordered basis for U . Given any vector $\mathbf{v} \in U$, there exists unique scalars $t_1, \dots, t_m$ such that

$$\mathbf{v} = t_1 \mathbf{b}_1 + \dots + t_m \mathbf{b}_m$$

and we define the

coordinate vector of a vector $\mathbf{v} \in U$ with respect to B to be the m -vector

$$C_B(\mathbf{v}) = \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix}.$$

See textbook figure fig-5-2-coordinate-basis-diagram for an illustration of this concept, taken from *Stanford’s MATH 51 Textbook*.

Note If $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ is an ordered basis for a subspace U , then $C_B(\mathbf{b}_i) = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th standard basis vector in $\mathbb{R}^m$.

Activity

Let $B = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ be an ordered basis for $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

What is $C_B(\mathbf{v})$ if

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix}.$$

We need to find scalars c_1, c_2 , and c_3 such that

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3.$$

Comparing entries of the vectors gives a system of three equations, i.e.,

$$c_1 + c_2 + 2c_3 = 1$$

$$c_2 = 1$$

$$c_3 = -5$$

Using back substitution, we find that $c_1 = 10$,

$c_2 = 1$, and $c_3 = -5$ are the unique solution to this system. Thus

$$C_B(\mathbf{v}) = \begin{bmatrix} 10 \\ 1 \\ -5 \end{bmatrix}.$$

Rank of a matrix

We are now going to connect the concept of dimension with the notion of rank. In this section, we will often realize vectors in $\mathbb{R}^n$ as rows rather than columns. The notion of span, linear independence, and basis are defined analogously as how they are defined for column vectors.

Definition: Column and Row Space Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

- The columns of A , considered as vectors in $\mathbb{R}^m$ span a *subspace* of $\mathbb{R}^m$ called the **column space** of A , denoted $\text{col}(A)$.
- The rows of A , considered as vectors in $\mathbb{R}^n$, span a *subspace* of $\mathbb{R}^n$ called the **row space** of A , denoted $\text{row}(A)$.

Note The column space is the image space from Subspaces written with a column-based name:

$$\text{col}(A) = \text{im}(A).$$

The notation $\text{im}(A)$ emphasizes outputs of the map $\mathbf{x} \mapsto A\mathbf{x}$. The notation $\text{col}(A)$ emphasizes the columns that span those outputs.

Lemma Let A and B be two row equivalent $m \times n$ matrices. Then $\text{row}(A) = \text{row}(B)$.

Proof Each time we apply a row operation to a matrix, we replace one or more of its rows with a linear combination of other rows. Thus the row space of a matrix A' obtained by applying a row operation to a matrix A must be contained in the row space of A . But applying the argument in reverse (since every row operation is reversible), the row space of A must be contained in the row space of A' . Thus the two row spaces are equal. But if the row space is not changed after applying a single row operation, it is not changed after applying an arbitrary number of row operations, and so the result follows.

Lemma If R is a matrix in row echelon form (recall Row echelon form and reduced row echelon form) then the non-zero rows of R are a basis of $\text{row}(R)$.

Proof The nonzero rows of R are linearly independent by the referenced item, and span $\text{row}(R)$ by definition. They thus by definition form a basis for $\text{row}(R)$.

Remark In Rank of a matrix, we defined the *rank* of a matrix as the number of leading 1s (i.e., the number of pivots) of any matrix R in row echelon form which is row-equivalent to A . By the referenced item, $\text{rank}(A) = \dim(\text{row}(A))$, which justifies that the rank of A is independent of the choice of matrix R .

Activity

Find a basis for $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \quad \mathbf{v}_4 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

The subspace V is the row space of the matrix

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}.$$

If we reduce this matrix to row echelon form, we thus find a basis for V :

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 1.5 & -3 \\ 0 & 1 & -2 \\ 0 & 0.5 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}$$

is a basis for V .

Theorem: Rank Theorem Let A denote any $m \times n$ matrix of rank r . Then

$$\dim(\text{col}(A)) = \dim(\text{row}(A)) = r.$$

Moreover, if A is row-equivalent to a matrix R in row echelon form, then

- The r leading rows of R are a basis for $\text{row}(A)$.
- If the pivots of R lie in columns $j_1, j_2, \dots, j_r$, then the columns $j_1, j_2, \dots, j_r$ of A are a basis for $\text{col}(A)$.

Warning: Pivot columns come from the original matrix Row reduction identifies pivot positions, but row operations do not preserve the original columns of A . When finding a basis for $\text{col}(A)$, use the pivot columns of the original matrix A , not the columns of the row-reduced matrix.

- **Goal:** basis for $\text{row}(A)$
Row reduction tells you: nonzero rows of REF/RREF
Use these vectors: rows of the reduced matrix
- **Goal:** basis for $\text{col}(A)$
Row reduction tells you: pivot column positions
Use these vectors: corresponding columns of original A

- **Goal:** basis for $\text{null}(A)$
Row reduction tells you: free variables
Use these vectors: basic solution vectors in $\mathbb{R}^n$
- **Goal:** rank
Row reduction tells you: number of pivots
Use these vectors: a number
- **Goal:** nullity
Row reduction tells you: number of free variables
Use these vectors: a number

Fact It follows from Rank Theorem that for any matrix A , $\text{rank}(A) = \text{rank}(A^T)$.

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
 What is $\dim(\text{row}(A))$?

The dimension of the row space of A is equal to the rank of A , which is 2 (the number of non-zero rows in the row echelon form for A , as computed in the solution to the activity).

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
Find a basis for the column space of A .

The leading 1s in the row echelon form for A computed in the activity occur in the 1st and 2nd columns, so the 1st and 2nd columns of A form a basis for the column space, i.e.,

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix} \right\}$$

is a basis for $\text{col}(A)$.

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.

What is $\dim(\text{col}(A))$?

The dimension of the column space of A is two, since we constructed a basis with two elements in the previous task. Alternatively, the dimension of the column space is equal to the rank of A , which we also computed to be two.

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
Find a basis for the nullspace of A .

To find a basis for the solutions to the equation $A\mathbf{x} = \mathbf{0}$, we use the row-echelon form computed in the activity and find basic solutions as in Basic solutions to homogeneous linear equations. The row-echelon form gives the equations $x_1 + \frac{1}{2}x_2 = 0$ and $x_2 - 2x_3 = 0$. Setting the free variable $x_3 = 1$, we obtain $x_2 = 2$ and $x_1 = -1$, so the basic solution is

$$\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

Thus $\left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right\}$ is a basis for the nullspace of A .

Definition The nullity of a matrix A is equal to $\dim(\text{null}(A))$. It is denoted $\text{nullity}(A)$.

Rank counts independent directions transmitted by a matrix. Nullity counts independent input directions forgotten by a matrix. The row-reduction definition of rank now connects to dimension.

Theorem: Rank-Nullity Theorem Let A denote an $m \times n$ matrix of rank r . Then

- The $n - r$ basic solutions to the equation $A\mathbf{x} = \mathbf{0}$ provided by the Gaussian algorithm are a basis for $\text{null}(A)$, so $\dim(\text{null}(A)) = n - r$.
- $\dim(\text{im}(A)) = r$.
- $\dim(\text{im}(A)) + \dim(\text{null}(A)) = n$.
- $\text{rank}(A) + \text{nullity}(A) = n$.

Note

input dimensions = transmitted dimensions + forgotten dimensions

$$n = \text{rank}(A) + \text{nullity}(A)$$

We can now state one of the preview patterns from Section 2.1, introduced in Two inputs, one output, in the language of null spaces.

Activity: Same output from a null-space direction

Suppose $A\mathbf{x} = \mathbf{y}$ and $A\mathbf{z} = \mathbf{0}$. Compute $A(\mathbf{x} + t\mathbf{z})$, where $t \in \mathbb{R}$. What does this say about uniqueness?

$$\begin{aligned}A(\mathbf{x} + t\mathbf{z}) &= A\mathbf{x} + tA\mathbf{z} \\&= \mathbf{y} + t\mathbf{0} \\&= \mathbf{y}.\end{aligned}$$

So every vector $\mathbf{x} + t\mathbf{z}$ gives the same output. A nonzero vector in the null space creates a whole line of inputs with the same output.

Activity: Revisiting the height map with rank and nullity

Return again to A map that forgets height. Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Find the column space, null space, rank, and nullity of P . How does rank-nullity describe the height direction that P forgets?

The column space is

$$\text{col}(P) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2,$$

so $\text{rank}(P) = 2$.

The null space is

$$\text{null}(P) = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\},$$

so $\text{nullity}(P) = 1$. Since P has three input coordinates,

$$\text{rank}(P) + \text{nullity}(P) = 2 + 1 = 3.$$

Rank-nullity says that the two output directions and the one forgotten height direction account for all three input directions.

Activity

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\text{im}(B)$.

We use the fact that $\text{im}(B) = \text{col}(B)$. Using Rank Theorem, since the RREF in the referenced item has pivots in the 1st and 3rd columns, the column space of B , and thus the image of B , has a basis given by the 1st and 3rd columns of B , i.e., we have a basis

$$\left\{ \begin{bmatrix} 3 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \end{bmatrix} \right\}.$$

Activity

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Find a basis for $\text{null}(B)$.

The 2nd and 4th columns of the matrix in the referenced item do not contain pivots, so x_2 and x_4 correspond to free variables in the equation, and solutions to $Ax = 0$ are specified by the two equations $x_1 = x_2/3 + 2x_4$ and $x_3 = -3x_4$. The two basis solutions give a basis for the nullspace, and thus we have a basis of the form

$$\left\{ \begin{bmatrix} 1/3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \end{bmatrix} \right\}.$$

Redundant features in a data matrix

A data matrix can have redundant columns. This means one feature column is a linear combination of other feature columns.

Rows below represent houses. The three features are

first-level area, second-level area, total area,

measured in hundreds of square feet:

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}.$$

Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of X . The third column satisfies

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2,$$

so the third feature carries no new independent direction. The dependence relation is

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0},$$

or equivalently

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

The null-space direction

$$\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

says that coefficient vectors differing by a multiple of $\mathbf{z}$ give the same prediction vector.

Activity: Redundant square-footage features

Use the data matrix

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

1. Compute $X\mathbf{z}$.
2. What does this say about the three feature columns?
3. Compare the coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}' = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Compute $X\mathbf{c}$ and $X\mathbf{c}'$.

4. The two coefficient vectors tell different stories. Why do they make the same predictions?
5. What warning does this give about asking which feature is most important?

First,

$$X\mathbf{z} = X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

This means the columns are dependent. The total-area column is the sum of the first-level and second-level columns.

Now,

$$X\mathbf{c} = \begin{bmatrix} 34 \\ 42 \\ 42 \\ 32 \end{bmatrix}, \quad X\mathbf{c}' = \begin{bmatrix} 34 \\ 42 \\ 42 \\ 32 \end{bmatrix}.$$

The coefficient vectors are different, but they differ by the null-space direction:

$$\mathbf{c} - \mathbf{c}' = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{z}.$$

Since $X\mathbf{z} = \mathbf{0}$,

$$X\mathbf{c} = X(\mathbf{c}' + \mathbf{z}) = X\mathbf{c}' + X\mathbf{z} = X\mathbf{c}'.$$

The predictions are the same, but the coefficient interpretations are different. One coefficient vector puts weight on first-level and second-level area separately. The other puts weight on first-level area and total area. With redundant features, individual coefficient importance is not uniquely determined by the data matrix.

Note: Common exact feature redundancies Common exact feature redundancies include:

- first-level area, second-level area, and total area;
- minutes and seconds, where seconds = 60 · minutes;
- height in inches and height in centimeters, where centimeters = 2.54 · inches;
- subtotal, tax, tip, and total bill, where total = subtotal + tax + tip;
- exam parts and exam total, where total = part 1 + part 2 + part 3.

These are linear redundancies. Not every relationship between features is linear. For example, rectangle area is width times height, not a linear combination of width and height.

Activity: Which features are linearly redundant?

For each feature list, decide whether one feature is a linear combination of the others.

1. first-level area, second-level area, total area;
2. height in inches, height in centimeters;
3. width, height, area of a rectangle;
4. quiz 1 score, quiz 2 score, quiz total;
5. bedrooms, bathrooms, selling price.

Items 1, 2, and 4 contain exact linear redundancies:

$\text{total area} = \text{first-level area} + \text{second-level area},$

$\text{centimeters} = 2.54(\text{inches}),$

and

$\text{quiz total} = \text{quiz 1} + \text{quiz 2}.$

Item 3 is related but not linearly redundant, because

$$\text{area} = (\text{width})(\text{height})$$

is not a linear combination of width and height.
Item 5 may contain useful statistical relationships, but there is no exact linear redundancy from the feature names alone.

Activity: Auditing redundant features in code

The matrix below stores the house features from the previous activity.

```
import numpy as np

X = np.array([
    [9, 7, 16],
    [11, 9, 20],
    [14, 0, 14],
    [8, 8, 16],
])

z = np.array([1, 1, -1])
c = np.array([3, 1, 0])
c_alt = np.array([2, 0, 1])

rank = int(np.linalg.matrix_rank(X))

X @ z, rank, X @ c, X @ c_alt
```

Here `c_alt` stores the vector c' .

Output:

```
(array([0, 0, 0, 0]), 2, array([34, 42, 42, 32]), array([34, 42, 42, 32]))
```

1. What does the zero output mean?
2. What does the rank say about the three feature columns?
3. Why do Xc and Xc' agree?
4. What does this say about interpreting coefficients when features are redundant?

The zero output means that $\mathbf{z}$ is a null-space direction. The rank is 2, so the three feature columns contain only two independent directions.

The vectors $\mathbf{c}$ and $\mathbf{c}'$ differ by $\mathbf{z}$, and $X\mathbf{z} = \mathbf{0}$. Therefore they give the same prediction vector. When features are redundant, different coefficient vectors can make the same predictions. Individual coefficients may not have a unique interpretation.

Activity: A difference matrix forgets level

Let

$$D = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

Compute $D \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$. What input direction is forgotten?

We get

$$D \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \mathbf{0}.$$

The matrix records consecutive differences. Adding the same constant to every entry does not change the differences, so the constant direction $(1, 1, 1, 1)$ is forgotten.

Activity: Reading the difference matrix in code

```
D = np.array([
    [-1, 1, 0, 0],
    [0, -1, 1, 0],
    [0, 0, -1, 1],
])

x = np.array([2, 5, 9, 10])
ones = np.ones(4, dtype=int)

D @ x, D @ (x + 10*ones), D @ ones
```

Output:

```
(array([3, 4, 1]), array([3, 4, 1]), array([0, 0, 0]))
```

1. What does $D @ x$ measure?
2. Why do the first two outputs agree?
3. What does $D @ \text{ones}$ say about the map?

The product $D @ x$ measures consecutive differences:

$$5 - 2 = 3, \quad 9 - 5 = 4, \quad 10 - 9 = 1.$$

Adding 10 to every entry changes the level but not the differences, so the first two outputs agree.

The output $D @ \text{ones}$ is 0 , so the constant direction is a null-space direction. The difference matrix forgets level.

Redundant features create null-space directions. If $Xz = 0$, then

$$X(c + tz) = Xc$$

for every scalar t . Thus many coefficient vectors can produce the same prediction vector. Rank counts how many independent feature directions remain; nullity counts how many independent coefficient changes are invisible to the data matrix.